

3D printing

Pad Printing is a process of printing images on 3D objects using a flexible rubber silicon pad to pick up the image from an inked plate and transfer it to the object

Vitraglyphic process

A team of engineers working at University of Washington have developed a technique to create a glass object using a normal 3D printer

Printer test

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Canon PIXMA iP3680

Setting up this printer is easy, but installing the ink tanks is pretty painful for first time users. Since the printer has five ink tanks, they are first installed on to a printer head, that, in turn, is placed in a specific groove. You then need to watch for a red light to indicate successful installation.

A good feature is the PictBridge port that will let you print photographs directly from camera memory cards. This is good for home use.

The printer is quieter than the PIXMA iP1980. It doesn't have the option of duplex printing, but instead prompts you to manually move the pages to the rear tray.

Performance

Under the magnifying glass, the Canon PIXMA iP1980 and iP3680 models had comparable quality. The iP3680 had better pronounced edges in the text print outs, as the iP1980 had some residue around the edges of the text in all text sizes. There was also fine blotting of ink seen. Printing times were also almost identical. In economy/speed modes, there's no difference in quality.

In the fast mixed prints, the iP3680 had visibly saturated colours. Speeds were pretty identical, again. There was a fine yellow noise all over the yellow text in black background areas. Greys are

noticeably lighter. The lines of text are heavily broken at smaller fonts.

The iP3680 has a little more detail in its photo prints. As for colours, the iP1980 showed more neutrality, but was still not able to produce complete accuracy. The iP3680, on the other hand was a little heavy on reds. In the highest photo quality setting, the iP3680 wins.

Mono laser printers – under Rs. 10,000

The printers here are targeted at small to medium size offices that have a lot of text to print. Since cost is an important factor, we split mono lasers into two categories – under Rs. 10,000 and over Rs. 10,000.

Canon Laser Shot LBP2900B

Canon's LBP2900B is a fast monochrome laser printer which can be used in homes as well as offices. It's quite attractive and very compact. To setup, all you have to do is slide the laser cartridge into position.

The software user interface for the LBP series is different from the PIXMA series, however navigation through the basic options is just as easy.

The paper feed tray located at the base is very strong, but the output tray,

though attractive, is quite weak. It might not be able to hold too many printer pages.

There's no duplex printing facility, which could have added a feather to its cap. On the whole, the LBP2900B is ideal for users who mainly print text documents, and is quite fast at spewing out a print or two. However, it suffers when you print, say, 15 pages. It took a whole minute and twenty two seconds to do the 15 page print test.

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Canon Laser Shot LBP3300B

HOW WE TESTED

The printers were all tested on a very basic Intel Pentium 4 system with 2 GB of RAM. All the printers were allowed to calibrate themselves via software, and a few test prints were taken before the actual testing began. Good bond quality A4 paper was used throughout the test.

Print quality was obviously one of the most important components of the test. Each of the prints were rated accordingly and compared to other printers of its category. Time was also recorded for every print taken. There were different prints taken. Since this isn't a photo printer test, we only had a basic photo print test while the major chunk

of the test was for text and documents with images in them. Neutral and natural colours was the key component along with accurate crisp prints. We also looked for flaws such as banding instead of smooth gradients, grain and uneven edges in text at different settings. Mixing of colours and uneven edges to text because of spreading ink was also looked at.

Features were also noted and weight given for each of them depending on their importance and the category of product. For example, a printer for SoHo or medium sized businesses should have network access, a cheap home printer need not. We looked at how easy it was

to replace the cartridges, toners and also how to restock the paper trays. Build quality of the printers and the multiple tray assembly was equally important. Of course, there were other things like noise, the printer software user interface, ease of setting up the printer on the network and many other finer aspects that were thoroughly explored and scored.

Not all printers require the same level of performance and quality. Each type of printer is made with a completely different purpose in mind. Different tests were given different weights for every category of printer. In the end, all of it taken into consideration and a final verdict given.